



Joint effects of several factors on cloud-to-ground lightning and rainfall in Nanning (China)

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ABSTRACT

The data of lightning flash density, total rainfall, surface temperature, surface humidity, convective available potential energy (CAPE) and particulate matter up to 10 μm in size (PM_{10}) over Nanning, China from July 2009 to December 2017 were studied on a yearly/monthly/hourly scale. It was found that the correlation between lightning and precipitation varies in various seasons, and the influencing factors of lightning/precipitation are different in various seasons. CAPE is the most important influencing factor of lightning on an hourly scale. Quantitative analysis showed the joint effects of surface temperature, surface humidity, CAPE and PM_{10} on lightning. For dry cases, lightning increases with increasing PM_{10} for the low value of PM_{10} , and decreases with increasing PM_{10} for the high value of PM_{10} . While the situation is the opposite for wet cases. It was also found that surface temperature and CAPE can affect the correlation between aerosol and lightning.

1. Introduction

Convective activity is one of the most fundamental meteorological events. It plays an important role in atmospheric energetics and hydrological cycle. Lightning activity associated to convective systems is a useful indicator of their rain yield (Hyun et al., 2010). The relation between lightning and convective rain has been shown by the study of visual observations (Battan, 1965), radar observations (Rutledge et al., 1992; Williams et al., 1992), lightning detection network observations (Kucińska et al., 2012; Strader and Ashley, 2014) and satellite observations (SenRoy and Balling Jr., 2013; Stolz et al., 2015). Lightning discharges require stronger and deeper uplift, whereas precipitation may occur even at weak and moderate updrafts (Siingh et al., 2015). Positive relation between lightning flash counts and rainfall has been reported in many regions, e.g., Taiwan Island (Liou and Kar, 2010), India (Siingh et al., 2014) and the Pacific coast of southern Mexico (Kucińska et al., 2012). Cloud-to-ground (CG) lightning can be used in local short term forecasting of heavy rain or in precipitation estimation (Siingh et al., 2014). However, the opposite behavior has been observed between rainfall and lightning flash counts on the global scale for tropical continental centers of rain and lightning range in the opposite order (Price, 2009; Siingh et al., 2013).

Convective activity is formed by buoyancy force controlled

primarily by surface air temperature; formation of convective activity is based on thermodynamic conditional instability of the boundary layer, the energy that feeds this process is convective available potential energy (CAPE) (Penki and Kamra, 2013). Higher surface temperature enhances surface heating. This excessive surface heating increases the magnitude of CAPE and its other characteristics in atmosphere, and then invigorates thunderstorm generation (Liou and Kar, 2010). With projections of a warmer climate in the future, how convective activity changes in a warmer world has become one of the key questions related to global warming (IPCC, 2007; Price, 2009).

Aerosol is another important role in regulating convection (Tan et al., 2016; Yuan et al., 2011). Increasing aerosols into atmosphere may reduce solar radiation through scattering, reflection and absorption; then atmospheric instability decreases, which reduces convective strength (Siingh et al., 2014; Siingh et al., 2015). On the other hand, moderate addition of aerosols in polluted regions may result in more cloud condensation nuclei (CCN) and cloud drops, which would invigorate convection (Rosenfeld et al., 2008). Besides, strong updraft, strong instability, greater cloud base height and high aerosol concentration can account for superlative liquid water content in the mixed-phase region, which may result in inverted polarity in thunderstorms; a large ice nuclei concentration may produce dominant positive charge in the lower portion of the mixed-phase region by maintaining

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ice saturation (Pawar et al., 2014). In general, aerosol effects can be divided into radiative and microphysical effects (Guo et al., 2016), and play an important role during convective activity forming process, which is mainly regulated by atmospheric humidity (Lal et al., 2018).

Lightning and precipitation, as well as their dependence on various parameters, have been studied by many studies (e.g., Ilotoviz et al., 2016; Lang et al., 2016; Pawar et al., 2017). Lightning and precipitation vary among different stations and among different seasons, while the influences of various factors on lightning/precipitation vary from station to station and from time to time (Singh et al., 2014; Tinmaker et al., 2016). Further analysis on more stations is needed for better understanding the relation between lightning and precipitation, as well as their dependence on various parameters.

Nanning city is the capital of Guangxi province, South China, which is a subtropical monsoon zone where convective events occur frequently causing one of the largest levels of precipitation and longest duration of flooding in China (Fu and Dan, 2014; Xu et al., 2009). Convective systems in this region are affected by geographical height, SST, tropical cyclones, the atmospheric wind field and the West Pacific Subtropical High, which has been explored by many studies (e.g. Liao et al., 2015; Yuan et al., 2014; Zhong and Chen, 2015). However, no paper, to our knowledge, has examined the relation between lightning and precipitation in Nanning, as well as its dependence on meteorological factors. Therefore, this paper focuses on Nanning and (1) studies the relation of lightning and precipitation; (2) investigates the effects of surface temperature, CAPE, particulate matter up to 10 μm in size (PM_{10}) and surface humidity on both lightning and precipitation; (3) analyses the interaction of these effects.

2. Dataset and methodology

Nanning city, the capital of Guangxi province, is the area studied (Fig. 1). It has an area of 6,476 km^2 with about 2.6 million inhabitants. The data, including meteorological data, lightning flashes data and PM_{10} concentration data, are from July 2009 to December 2017.

The meteorological data, including total rainfall, surface temperature, surface humidity and CAPE data, were obtained from Nanning weather station (longitude 108.35°E, latitude 22.82°N, altitude 121.6 m, World Meteorological Organization ID code: 59431). It has a subtropical monsoon climate with an annual average temperature of 21.8 °C, an annual rainfall of 1,310 mm, and an annual sunshine duration of 1,585 h. The data of surface temperature, rainfall and surface humidity were observed at surface once an hour. The data of

CAPE was based on the sounding data and was calculated pseudo adiabatically from temperature and relative humidity fields on 8 vertical pressure levels (1000/925/850/700/500/400/300/200 mb), which was observed only at 08 and 20 LST (LST = UTC + 8 h), while hourly CAPE in other hours was calculated by interpolation using the data in 08 and 20 LST.

The hourly lightning flashes data were obtained from Guangxi meteorological bureau. It was observed by a local lightning monitoring system. This represents a province-wide lightning detection (LD) network consisting of 11 lightning sensors to provide the coverage of lightning flashes for the entire Guangxi province (Fig. 1). The CG lightning detection efficiency and median error of the location accuracy are above 90% and < 1 km, respectively. It uses the Improved Performance through Combined Technology (IMPACT) method, which combines the direction finding and time of arrival technology to determine the lightning location. A spatial scale of 0.1° latitude $\times$ 0.1° longitude (about 100 km^2) surrounding the weather station was selected to find out the CG lightning flash counts following Liou and Kar (2010). Lightning flash density, which is defined as the number of flashes, is used to describe lightning activity. Following the previous papers (e.g., Zheng et al., 2016), + CG lightning with current < 0 kA was removed from the dataset because it could be a misinterpretation of cloud lightning.

The daily PM_{10} concentration data were obtained from Nanning environmental protection bureau. It was reported that PM_{10} concentration is better than aerosol optical depth (AOD) for studying the aerosol effect on precipitation because the former could be observed under all-sky conditions while the latter is measurable only under cloud-free conditions and gives no information about where aerosols reside in the vertical column; besides, PM_{10} has a close relation with $\text{PM}_{2.5}$ (particulate matter up to 2.5 μm in size) and is sufficient for studying the radiative and microphysical effects of aerosols on convective systems over South China (Guo et al., 2016). Thus, ground measurement of PM_{10} concentration data rather than AOD data was chosen as a proxy for CCN in this study.

The data have undergone quality control checks at Guangxi meteorological bureau and Nanning environmental protection bureau. Data availability is above 99.5% for the variables total CG flashes, + CG flashes, rainfall, surface temperature, surface humidity and CAPE, and is above 97% for the variable PM_{10} concentration.

Following the previous papers (Liou and Kar, 2010; Tan et al., 2016), the parameters P_{+CG} and rain yield were computed as following.

$$P_{+CG} = \frac{\text{number of positive CG flash counts during a certain time period}}{\text{number of total CG flash counts over the same time period}} \quad (1)$$

$$\text{Rain yield} = \frac{\text{rainfall during a certain time period}}{\text{number of total CG flash counts over the same time period}} \quad (2)$$

where “a certain time period” is a year, a month or an hour respectively. However, rain yield will not be analyzed on an hourly scale for a significant lag was reported between diurnal cycles of lightning and rainfall (Price, 2013).

3. Results

3.1. Annual variation

The annual variations of total CG flashes, P_{+CG} , total rain, rain yield, surface temperature, CAPE, PM_{10} and surface humidity are presented in Fig. 2. When surface temperature decreases from 2010 to 2011, total CG flashes decreases at the same time. Moreover, both of them increase from 2011 until reaching peaks in 2016, and then decrease in 2017. The trends of CAPE and surface humidity are similar to those of surface temperature/total CG flashes except that peak is in

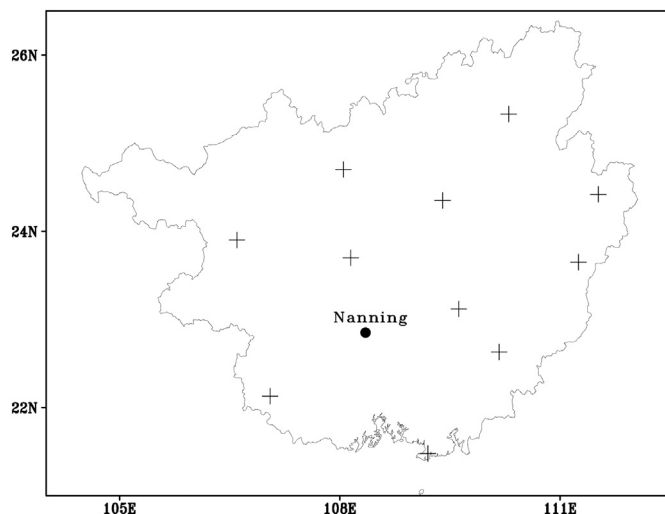


Fig. 1. Map of Guangxi province, China showing the study region for the present study: Nanning. The locations of 11 lightning sensors are shown by plus signs.

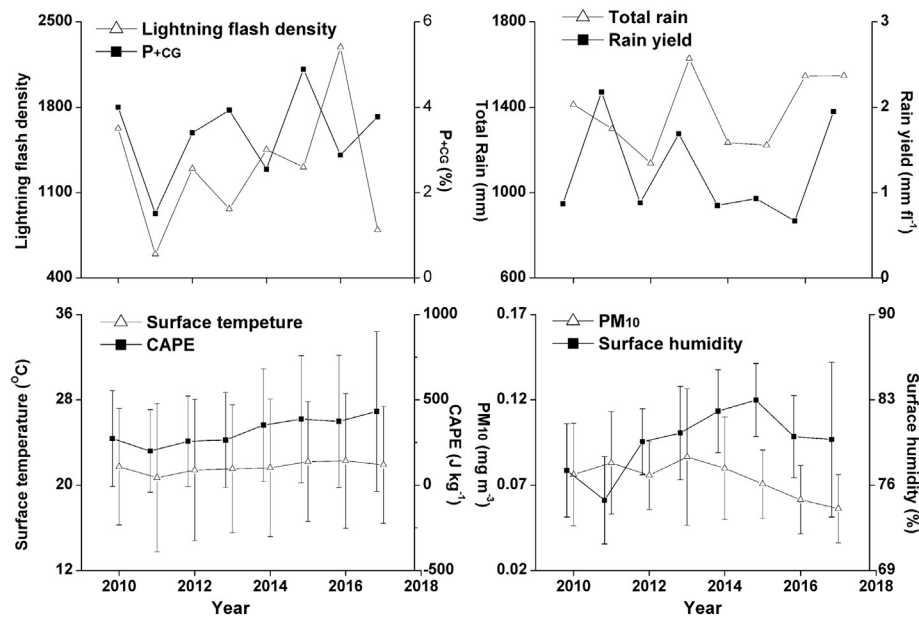


Fig. 2. Annual lightning flash density, P_{+CG} , total rain, rain yield, surface temperature, CAPE, PM_{10} and surface humidity. Vertical error bars show the standard deviations.

2017 for CAPE, and is 2015 for surface humidity.

PM_{10} vibrates during 2010–2012 and then decreases gradually from 2013 to 2017 due to less pollutant emissions and strict environmental controls in recent years, and might have a relation with the variation of the east Asian monsoon circulation (Zhao et al., 2017). The lowest value of PM_{10} is found in 2017, which might play a role in the dramatic decrease of total CG from 2016 to 2017.

P_{+CG} has a similar trend to those of total CG flashes, surface temperature, CAPE and surface humidity during 2010–2012. While during 2013–2017, when total CG flashes, surface temperature, CAPE and surface humidity show an increasing trend, vibration is found in P_{+CG} . This might be attributed to the decrease trend of PM_{10} for aerosol has a positive relation with P_{+CG} (Pawar et al., 2014; Tan et al., 2016).

Total rain vibrates during 2010–2017, which is not consistent with other parameters. Since no increasing or decreasing trend is found for total rain, it is expected that rain yield, which is ratio between total rain and total CG flashes, has a reverse trend with total CG flashes. And thus rain yield increases from 2010 to 2011, then decreases until 2016, and increases again from 2016 to 2017.

In brief, the annual data analysis partly agrees with the findings reported by Steiger et al. (2002) in Houston, USA, Kar et al. (2009) in South Korea, Tan et al. (2016) in Nanjing, China, Lal et al. (2013), Lal and Pawar (2011) in India. Considering that the sample number of the yearly data is low, further analysis on seasonal and diurnal variations is needed.

3.2. Seasonal variation

The seasonal variation based on monthly date is studied. It is shown from Fig. 3 that all the parameters have obvious seasonal variations except surface humidity. The parameters lightning flash density, total rain, surface temperature and CAPE are high in May–September and are low in October–April, which is similar to the results reported in other areas, e.g., India (Siingh et al., 2015), Poland (Taszarek et al., 2015) and Korea (Hyun et al., 2010). The seasonal variations of both lightning flash density and total rain are attributed to the variations of surface temperature and CAPE (Siingh et al., 2014). All of these four parameters have only one peak. Both lightning flash density and CAPE show maximums in June, whereas both total rain and surface temperature show maximums in July.

On the contrary, PM_{10} , P_{+CG} and rain yield are high in October–April, and are low in May–September, which agrees with the results reported in many other areas, e.g., Poland (Taszarek et al., 2015), the contiguous United States (Orville, 1997), Taiwan Island (Liou and Kar, 2010). The relative low value of PM_{10} in May–September is attributed to unstable atmospheric conditions and rainy weather that are favorable for the dispersion and scavenging of airborne particulate matter (Wan et al., 2010), which could be seen from the high values of CAPE and total rain in these months. The higher value of rain yield in October–April is mainly due to the comparatively lower number of lightning flash density than that in May–September (Liou and Kar, 2010). Besides, the seasonal variation of P_{+CG} is explained by different types of precipitation systems. Precipitation in the study area during October–April is mainly related either to tropical cyclones or to transient disturbances embedded in the northeasterly monsoon flow during the passage of cold fronts (Wang et al., 2011), since these types of precipitation systems carry less lightning (Liou and Kar, 2010), the rain yield in October–April is comparatively higher than in May–September.

It is evident from Fig. 3 that the values of the parameters are quite different in various seasons. For studying the correlation of these parameters in different seasons, each year is divided into dry season, premonsoon season and monsoon season. Following Liao et al. (2015) and Lal and Pawar (2009), the premonsoon season is defined from April to June, the monsoon season is defined from July to September, while the dry season is defined from October to March of the following year. The Pearson correlation among these parameters is calculated and the results are presented in Tables 1–3.

It is shown that lightning flash density has a better positive correlation with total rainfall during premonsoon seasons ($R = 0.622$, $P = .001$, see in Table 1) than that of monsoon seasons ($R = 0.401$, $P = .038$, see in Table 2). Whereas the correlation between these two parameters is not significant during dry seasons ($R = 0.245$, $P = .083 > 0.05$, see in Table 3). A relation is not significant if P is above 0.05. Besides, lightning flash density significantly correlates with rain yield, surface temperature, CAPE and PM_{10} during the premonsoon seasons, whereas these relations are not significant in both monsoon and dry seasons except that a positive correlation between lightning flash density and CAPE is significant for dry seasons ($R = 0.544$, $P < .001$). The different correlations of lightning flash density in various seasons is mainly caused by the different cloud base

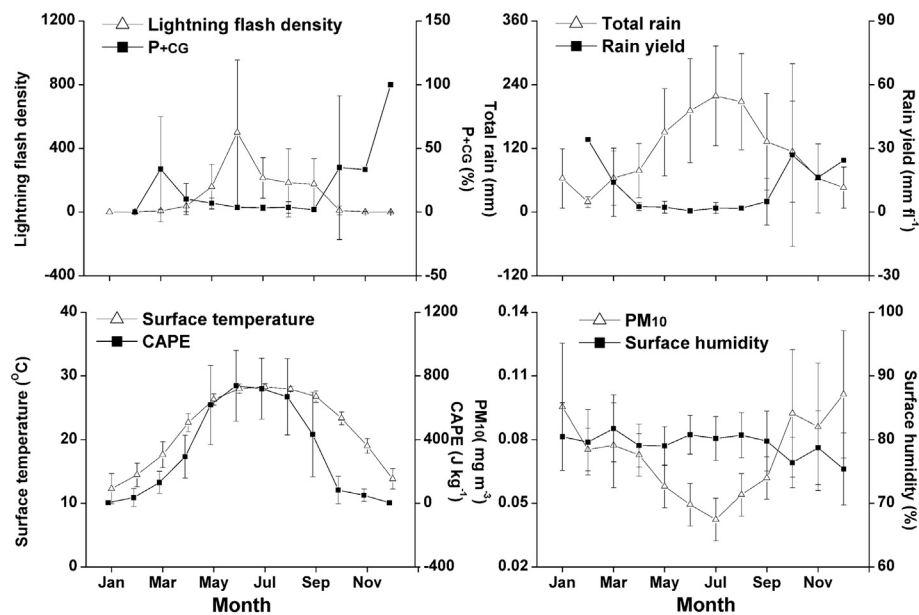


Fig. 3. Monthly variation of lightning flash density, P_{+CG} , total rain, rain yield, surface temperature, CAPE, PM_{10} and surface humidity. Vertical error bars show the standard deviations.

height and aerosol concentration in various seasons, which has been pointed out by Lal and Pawar (2009).

P_{+CG} has a significant positive correlation with rain yield during premonsoon seasons ($R = 0.632$, $P = .002$, see in Table 1), which is attributed to that convections with +CG lightning are usually stronger than those with -CG lightning (Coquillat et al., 2013) and bring more rainfall in southern China (Guo et al., 2016). The reason why this correlation is significant in premonsoon seasons rather than monsoon seasons is that heavy convective activity is often in intrusions of frontal systems relating lightning during premonsoon seasons whereas during monsoon seasons they are often in typhoons (Wang et al., 2011). The difference of heavy convective activity formation is also used to explain the significant correlation between P_{+CG} and surface temperature in premonsoon seasons ($R = -0.539$, $P = .01$, see Table 1) rather than in monsoon seasons. Besides, during dry seasons, P_{+CG} has significant correlations with total rain and PM_{10} ($R = -0.655$, $P = .029$; $R = 0.8$, $P = .003$ respectively, see Table 3). The correlation between P_{+CG} and PM_{10} is attributed to that elevated aerosols may enhance +CG activity (Pawar et al., 2014; Tan et al., 2016). This significant correlation is found during dry seasons rather than during premonsoon/monsoon seasons, and is attributed to the high values of PM_{10} during dry seasons (see Fig. 3).

Total rain has significant correlations with both surface temperature and PM_{10} ($R = 0.44$, $P = .031$; $R = -0.539$, $P = .007$ respectively, see in Table 1) during premonsoon seasons rather than monsoon seasons, which is similar to that of P_{+CG} . It is attributed, as that of P_{+CG} , to different heavy precipitation formations between premonsoon and

monsoon seasons (Wang et al., 2011). Besides, total rain has a significant correlation with surface temperature during dry seasons ($R = 0.328$, $P = .019$, see Table 3), which might have a relation with anomalies of sea surface temperature relating precipitation formation (Cai and Qian, 2012).

Rain yield has negative correlations with surface temperature and CAPE during premonsoon seasons ($R = -0.563$, $P = .006$; $R = -0.652$, $P = .001$ respectively, see in Table 1), and has a negative correlation with surface temperature during monsoon seasons ($R = -0.47$, $P = .013$, see in Table 2), i.e., rain yield decreases when surface temperature/CAPE increases. This phenomenon is attributed to that increasing surface temperature and CAPE promotes atmospheric instability and stronger air motion that helps in thunderstorm generation (Liou and Kar, 2010).

3.3. Diurnal variation

Hourly data is calculated for studying diurnal variation in this section. Hourly PM_{10} is not available and thus daily PM_{10} is used as a surrogate. For diurnal lightning statistics, data are used only for convective days (i.e., daily lightning flash density is above zero). While for total rainfall, in order to accurately characterize precipitation variation, only the rainy days are selected. Besides, for studying the diurnal variation of total rainfall, the data are grouped into three clusters, i.e., the rainfall cluster with +CG lightning, the rainfall cluster with CG lightning and the rainfall cluster without lightning; if lightning is observed during a day (from 0 LST through 24 LST), the 24 times of hourly

Table 1

The correlation coefficient (R) and significance level (P) among monthly lightning flash density, P_{+CG} , total rain, rain yield, surface temperature (T), CAPE, PM_{10} and surface humidity for premonsoon seasons (April, May and June).

		Flashes	P_{+CG}	Total rain	Rain yield	T	CAPE	PM_{10}	Humidity
Flashes	R	–	–0.291	0.622	–0.440	0.567	0.595	–0.530	0.166
	P	–	0.189	0.001	0.040	0.004	0.002	0.008	0.438
P_{+CG}	R	–0.291	–	0.029	0.632	–0.539	–0.417	0.308	0.049
	P	0.189	–	0.899	0.002	0.010	0.054	0.163	0.829
Total rain	R	0.622	0.029	–	–0.135	0.440	0.311	–0.539	0.101
	P	0.001	0.899	–	0.548	0.031	0.139	0.007	0.639
Rain yield	R	–0.440	0.632	–0.135	–	–0.563	–0.652	0.306	–0.398
	P	0.040	0.002	0.548	–	0.006	0.001	0.167	0.066

Table 2

The same as Table 1, but for monsoon seasons (July, August and September).

		Flashes	P _{+CG}	Total rain	Rain yield	T	CAPE	PM ₁₀	Humidity
Flashes	R	–	–0.278	0.401	–0.352	0.141	0.012	0.126	–0.054
	P	–	0.160	0.038	0.072	0.483	0.951	0.532	0.789
P _{+CG}	R	–0.278	–	–0.107	–0.092	0.206	0.082	–0.258	0.195
	P	0.160	–	0.595	0.648	0.302	0.685	0.193	0.329
Total rain	R	0.401	–0.107	–	–0.054	0.204	0.377	–0.181	0.329
	P	0.038	0.595	–	0.789	0.307	0.053	0.366	0.093
Rain yield	R	–0.352	–0.092	–0.054	–	–0.470	–0.249	0.340	0.200
	P	0.072	0.648	0.789	–	0.013	0.210	0.083	0.316

measurement in this day are included in the rainfall cluster with lightning.

It is shown in Fig. 4a that the distribution of CG lightning is similar to that of +CG lightning. Both of these two parameters have maximums from afternoon to early evening. It is a typical variation found in the other regions of the world, and is associated with the maximum air temperature and convective activity during this period caused by the diurnal cycle of insolation (Xie et al., 2013). Secondary peaks are found during early morning, which is connected with the strengthening convection caused by nocturnal cooling of clouds (Chen et al., 2009). While for P_{+CG}, it is the highest during 8–9 LST when +CG flashes is relatively high while total CG flashes is relatively low. Besides, the lowest value of P_{+CG} is found in 22 LST when +CG flashes is relatively low and total CG flashes is relatively high.

The rain cluster with CG lightning has a similar diurnal variation as that of +CG lightning (Fig. 4b). For both of these two clusters, rainfall is obvious higher after midday than that before midday, which is similar to the distribution of lightning. The diurnal variations of lightning and rainfall corroborate the previous finding of Guo et al. (2016), Xu and Zipser (2011), Yu et al. (2007). These studies have found that both lightning flash density and precipitation have afternoon peaks in southern China. Surface temperature often reaches a maximum in the afternoon and the atmosphere in the lower troposphere tends to be unstable in southern China, and thus the atmospheric profile is favorable for the development of convective activity and lightning (Guo et al., 2016). However, for the rainfall cluster without lightning, no afternoon peak is found because the precipitation systems of this kind are not generated by convective activity.

The variations of the three clusters in Fig. 3b are compared with each other. Rainfall without lightning is obviously lower than those of having lightning (including the ones with CG lightning and the ones with +CG lightning), which suggests that most precipitation in the study area is generated by convective activity. Besides, rainfall is higher for the ones with +CG lightning than for the ones with CG lightning in most hours, which is attributed to that stronger convections of +CG lightning bringing more rainfall when comparing with –CG lightning in southern China (Coquillat et al., 2013; Guo et al., 2016).

It was reported that whether lightning activity increases in a warmer world is under debate (Price, 2009). A comparison of lightning between cold and warm conditions might provide some information about this question. To address it, the data are grouped into two clusters

Table 3

The same as Table 1, but for dry seasons (from October to March of the following year).

		Flashes	P _{+CG}	Total rain	Rain yield	T	CAPE	PM ₁₀	Humidity
Flashes	R	–	–0.326	0.245	–0.390	0.273	0.544	–0.202	0.022
	P	–	0.328	0.083	0.235	0.053	< 0.001	0.156	0.879
P _{+CG}	R	–0.326	–	–0.655	–0.392	–0.122	–0.315	0.800	–0.521
	P	0.328	–	0.029	0.232	0.722	0.346	0.003	0.100
Total rain	R	0.245	–0.655	–	0.226	0.328	0.138	–0.232	0.227
	P	0.083	0.029	–	0.504	0.019	0.336	0.102	0.108
Rain yield	R	–0.390	–0.392	0.226	–	0.127	–0.229	–0.275	0.061
	P	0.235	0.232	0.504	–	0.711	0.499	0.413	0.859

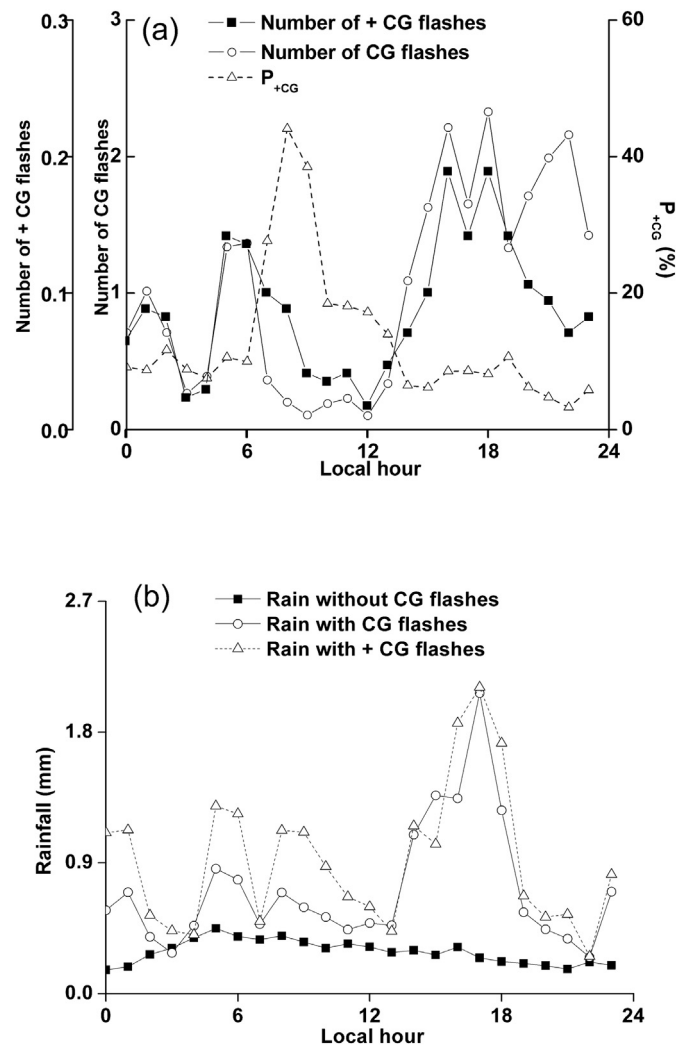


Fig. 4. Diurnal variations of the hourly (a) lightning flash density, only the data with daily lightning flash density above zero are selected; (b) rainfall, only the rainy days are selected.

Table 4

The correlation coefficient (R) and significance level (P) between dependent (hourly lightning flash density) and independents (hourly surface temperature (T)/CAPE/PM₁₀/humidity) for different clusters. Sample Number (N) for each cluster is presented. The data are grouped by hourly T/CAPE/PM₁₀/humidity respectively. Hourly PM₁₀ is not available and thus daily PM₁₀ is used as a surrogate. Only the data with daily lightning flash density above zero are selected.

Data cluster	N		Independent			
			T	CAPE	PM ₁₀	Humidity
T < 27 °C	5300	R	0.013	0.085	−0.004	0.011
		P	0.361	< 0.001	0.772	0.425
T > 27 °C	4874	R	−0.028	0.036	0.025	0.028
		P	0.052	0.012	0.081	0.047
CAPE < 750 J kg ^{−1}	5079	R	0.002	0.040	0.004	0.034
		P	0.861	0.004	0.800	0.017
CAPE > 750 J kg ^{−1}	5095	R	−0.061	0.072	0.008	0.030
		P	< 0.001	< 0.001	0.582	0.030
PM ₁₀ < 0.042 mg m ^{−3}	5095	R	−0.027	0.032	< 0.001	0.044
		P	0.054	0.024	0.989	0.002
PM ₁₀ > 0.042 mg m ^{−3}	5079	R	−0.013	0.069	0.008	0.021
		P	0.366	< 0.001	0.559	0.132
Humidity < 87%	5175	R	−0.009	0.085	0.028	0.023
		P	0.541	< 0.001	0.048	0.094
Humidity > 87%	4999	R	−0.003	0.027	−0.007	−0.001
		P	0.859	0.053	0.614	0.966

by different hourly surface temperature. Data are used only for convective days, i.e., if lightning is observed during a day (from 0 LST through 24 LST), the 24 times of hourly measurement in this day are included for analyzing. The value 27 °C is used as a threshold to ensure that the sample numbers of the two clusters are roughly equal, i.e., the sample number is 5300 for the cluster surface temperature (T) < 27 °C, and is 4874 for the cluster T > 27 °C (see Table 4). Then the dependence of lightning on surface temperature/CAPE/PM₁₀/surface humidity is analyzed on an hourly scale for these two clusters. The results presented in Table 4 show that CAPE positively affects lightning flash density in the two clusters, and is more obvious for the cluster T < 27 °C (R = 0.085, P < .001) than that of T > 27 °C (R = 0.036, P = .012). Besides, surface humidity positively affects lightning flash density for the cluster T > 27 °C (R = 0.028, P = .047) rather than that of T < 27 °C.

Following that of surface temperature, the data are grouped by CAPE, PM₁₀ and surface humidity respectively. For CAPE the threshold is 750 J kg^{−1}, for PM₁₀ it is 0.042 mg m^{−3}, while for surface humidity 87% is a threshold. Sample number for each cluster is presented in Table 4. When the data are grouped by CAPE, it is found that CAPE has a more obvious effect for the cluster CAPE > 750 J kg^{−1} (R = 0.072, P < .001) than that of CAPE < 750 J kg^{−1} (R = 0.04, P = .004), whereas surface humidity has a more obvious effect for the cluster CAPE < 750 J kg^{−1} (R = 0.034, P = .017) than that of CAPE > 750 J kg^{−1} (R = 0.03, P = .03). Besides, surface temperature affects lightning for the cluster CAPE > 750 J kg^{−1} (R = −0.061, P < .001) rather than that of CAPE < 750 J kg^{−1}.

When the data are grouped by PM₁₀, it is found that CAPE has a stronger positive effect on lightning when under polluted condition (R = 0.069, P < .001) than under clean condition (R = 0.032, P = .024). Besides, surface humidity positively affects lightning under clean condition (R = 0.044, P = .002).

Then, the data grouped by surface humidity are analyzed. Significant correlation is found for both CAPE and PM₁₀, but only in dry condition rather than wet condition, i.e., CAPE positively affects lightning with R = 0.085, P < .001, PM₁₀ positively affects lightning with R = 0.028, P = .048.

For studying the joint effect of the four parameters (surface temperature, CAPE, PM₁₀ and surface humidity) on lightning, the data are

grouped into 16 clusters by the four parameters. Diurnal variations of the 16 clusters are presented in Fig. 5. The hourly values are zero or nearly zero for most hours. However, difference is observed in the peaks for the clusters. The variations of dry conditions are presented in Fig. 5a–d, in which peak is higher for pollution cases than for clean cases, i.e., it is higher in line × 4 than in line × 3, higher in line × 6 than in line × 5, higher in line × 8 than in line × 7. An exception is found in Fig. 5a where the peak of line × 1 is a little higher than that of line × 2. However, the peak of line × 1 is very low in wet conditions, especially when compared with that of line × 4.

When the cases of wet conditions are analyzed, a reverse result is found (Fig. 5e–h). It is shown that peak is higher for clean cases than for pollution cases, i.e., it is higher in line × 13 than in line × 14, higher in line × 11 than in line × 12. The peak of line × 9 is roughly equal to that of line × 10. An exception is found in Fig. 5h where the peak of × 16 is higher than that of line × 15. However, the peak of × 16 is relatively low when compared with that of line × 13.

For the 16 clusters in Fig. 5, the highest peak is found in line × 13, i.e., the optimal conditions for lightning is high surface temperature, low CAPE, low PM₁₀ and high surface humidity. The second highest peak is found in line × 4, i.e., the second optimal condition is low surface temperature, high CAPE, high PM₁₀ and low surface humidity. Besides, the values of lines × 11 and × 16 are also relatively high.

For quantifying the joint effect of PM₁₀ and surface humidity on lightning, the data are grouped by hourly surface humidity, then hourly lightning flash density is calculated as a function of daily PM₁₀ following Rosenfeld et al. (2007). Line Y1 in Fig. 6a shows lightning varying with increasing PM₁₀ for dry cases (surface humidity < 87%). When PM₁₀ increases from low value, lightning increases gradually until the value of PM₁₀ reaches 0.115 mg m^{−3}, thereafter it decreases with increasing PM₁₀. For the segment PM₁₀ < 0.115 mg m^{−3} and the segment PM₁₀ > 0.115 mg m^{−3} of line Y1, the linear regression line and regression equation are analyzed respectively (not shown in Fig. 6), the regression coefficients (b) of regression equations are presented in Table 5. The b value of Y1 is 0.05 for the segment PM₁₀ < 0.115 mg m^{−3}, that is to say, the number of total CG flashes increases 0.05 when PM₁₀ increases 1 mg m^{−3}. While for the segment PM₁₀ > 0.115 mg m^{−3} of line Y1, it is shown that the number of total CG flashes decreases 0.017 when PM₁₀ increases 1 mg m^{−3}.

Lightning varying with increasing PM₁₀ for wet cases (surface humidity > 87%) is also analyzed and the result is presented as line Y2 in Fig. 6a and in Table 5. Contrary to line Y1, lightning decreases with increasing PM₁₀ when PM₁₀ < 0.085 mg m^{−3} (b = −0.041), and increases with increasing PM₁₀ when PM₁₀ > 0.085 mg m^{−3} (b = 0.148).

Then, CAPE effect is analyzed based on the result of Fig. 6a. The data are grouped into 4 clusters by surface humidity and CAPE. The result is presented in Fig. 6b and Table 5. The two dry clusters are presented as lines Y3 and Y4, which show that lightning increases with increasing PM₁₀ for the low value of PM₁₀, and decreases with increasing PM₁₀ for the high value of PM₁₀. While a reverse trend is found for the wet cluster humidity > 87%, CAPE > 750 J kg^{−1} (shown as line Y6), lightning decreases with increasing PM₁₀ for the low value of PM₁₀, and increases with increasing PM₁₀ for the high value of PM₁₀. Besides, for another wet cluster humidity > 87%, CAPE < 750 J kg^{−1} (shown as line Y5), lightning decreases with increasing PM₁₀ during the whole range of PM₁₀.

When vibration amplitudes are compared between low and high CAPE clusters, it is found that vibration is stronger in high CAPE clusters than in low CAPE clusters, i.e., variation is more dramatic in line Y4 than in line Y3, more dramatic in line Y6 than in line Y5.

Next, surface temperature effect is analyzed based on the result of Fig. 6b. The data are grouped into 8 clusters by surface humidity, CAPE and surface temperature. The result is presented in Fig. 6c,d and Table 5. It is found that for the dry clusters (lines Y7, Y8, Y11, Y12), lightning follows PM₁₀ up to when PM₁₀ is low, thereafter lightning

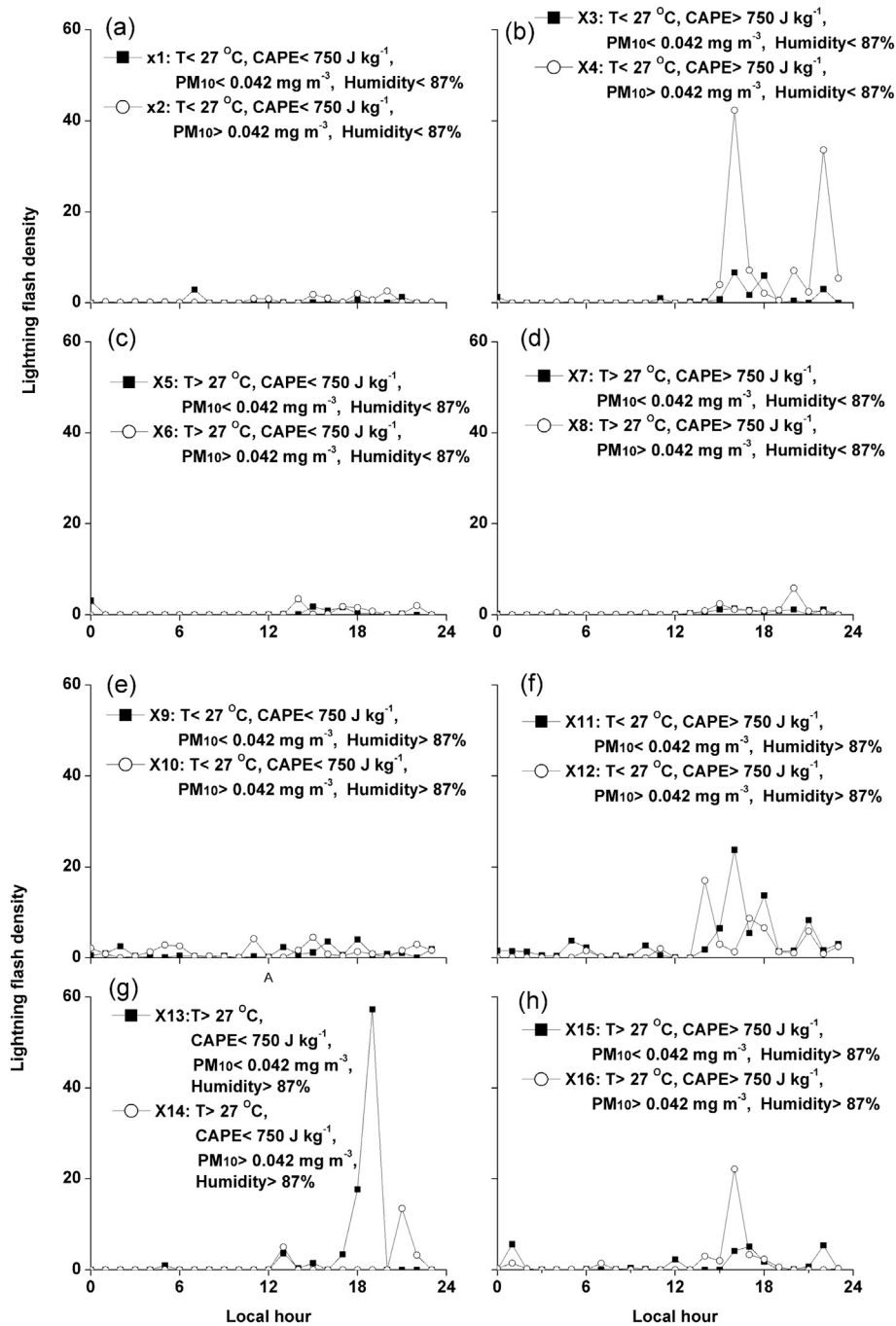


Fig. 5. Diurnal variations of the hourly lightning flash density for different clusters. The data are grouped by hourly surface temperature (T)/CAPE/PM₁₀/humidity. Hourly PM₁₀ is not available and thus daily PM₁₀ is used as a surrogate. Only the data with daily lightning flash density above zero are selected.

decreases. However, a reverse trend is found for the wet clusters (lines Y9, Y10, Y13, Y14), i.e., lightning decreases with increasing PM₁₀ for the low value of PM₁₀, and increases with increasing PM₁₀ for the high value of PM₁₀.

When vibration amplitudes are compared among the four low surface temperature clusters (lines Y7, Y8, Y9, Y10 of Fig. 6c), it is shown that the strongest vibration is found in line Y8 (humidity $< 87\%$, $\text{CAPE} > 750 \text{ J kg}^{-1}$, $T < 27^{\circ}\text{C}$), that is, the number of total CG flashes increases 0.331 when PM₁₀ increases 1 mg m^{-3} if $\text{PM}_{10} < 0.07 \text{ mg m}^{-3}$, and decreases 0.254 when PM₁₀ increases 1 mg m^{-3} if $\text{PM}_{10} > 0.07 \text{ mg m}^{-3}$. While among the four high surface temperature clusters (lines Y11, Y12, Y13, Y14 of Fig. 6d), vibration is the strongest in line Y13 (humidity $> 87\%$, $\text{CAPE} < 750 \text{ J kg}^{-1}$,

$T > 27^{\circ}\text{C}$), that is, the number of total CG flashes decreases 0.176 when PM₁₀ increases 1 mg m^{-3} if $\text{PM}_{10} < 0.07 \text{ mg m}^{-3}$, and increases 0.346 when PM₁₀ increases 1 mg m^{-3} if $\text{PM}_{10} > 0.07 \text{ mg m}^{-3}$.

4. Discussion

The results in Table 4 show that CAPE has the most significant correlation among the four independents for most clusters, which suggests that CAPE is the key influencing factor regulating lightning on an hourly scale. The correlations between CAPE and lightning in all clusters are significant except that for the cluster humidity $> 87\%$ the significant level p is 0.053, which is a little higher above 0.05. The results about CAPE are in line with those observed in previous studies

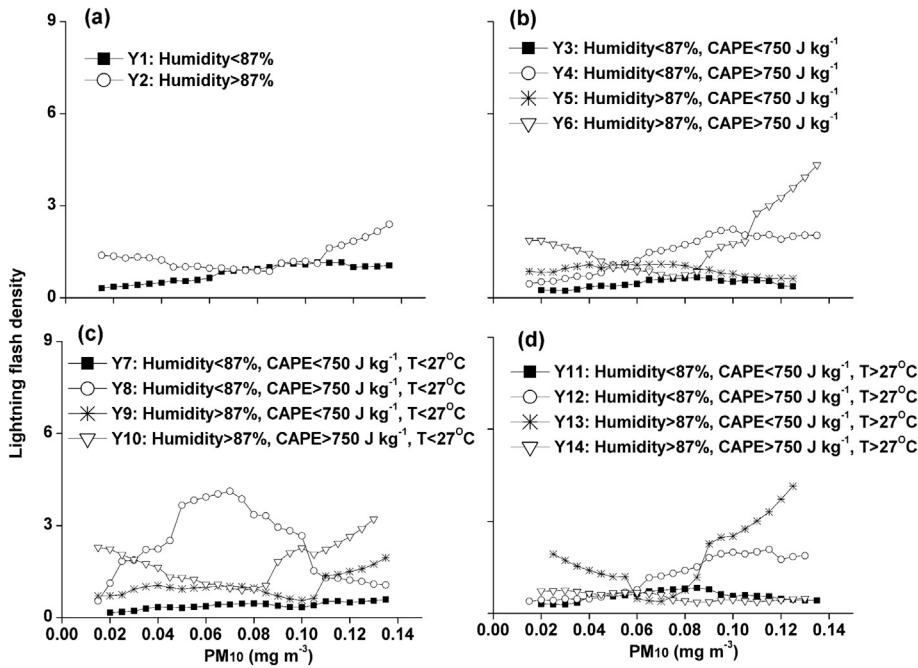


Fig. 6. Hourly lightning flash density as a function of daily PM_{10} for different clusters. The data are grouped by hourly surface temperature (T)/CAPE/humidity. The lines are interpolated curve fits with a weight applied. The weight is applied to 20% of the data (-10% to $+10\%$ of the data around the current point). Only the data with daily lightning flash density above zero are selected.

Table 5

The regression coefficients (b) of linear regression line for lines in Fig. 6.

Description of line	Range of PM_{10}	b
Y1: Humidity < 87%	< 0.115 $mg\ m^{-3}$ > 0.115 $mg\ m^{-3}$	0.050 −0.017
Y2: Humidity > 87%	< 0.085 $mg\ m^{-3}$ > 0.085 $mg\ m^{-3}$	−0.041 0.148
Y3: Humidity < 87%, CAPE < 750 $J\ kg^{-1}$	< 0.085 $mg\ m^{-3}$ > 0.085 $mg\ m^{-3}$	0.037 −0.032
Y4: Humidity < 87%, CAPE > 750 $J\ kg^{-1}$	< 0.100 $mg\ m^{-3}$ > 0.100 $mg\ m^{-3}$	0.112 −0.018
Y5: Humidity > 87%, CAPE < 750 $J\ kg^{-1}$	All	−0.015
Y6: Humidity > 87%, CAPE > 750 $J\ kg^{-1}$	< 0.075 $mg\ m^{-3}$ > 0.075 $mg\ m^{-3}$	−0.111 0.330
Y7: Humidity < 87%, CAPE < 750 $J\ kg^{-1}$, T < 27 °C	< 0.085 $mg\ m^{-3}$ 0.08–0.1 $mg\ m^{-3}$ > 0.100 $mg\ m^{-3}$	0.025 −0.035 0.030
Y8: Humidity < 87%, CAPE > 750 $J\ kg^{-1}$, T < 27 °C	< 0.070 $mg\ m^{-3}$ > 0.070 $mg\ m^{-3}$	0.331 −0.254
Y9: Humidity > 87%, CAPE < 750 $J\ kg^{-1}$, T < 27 °C	< 0.100 $mg\ m^{-3}$ > 0.100 $mg\ m^{-3}$	−0.006 0.175
Y10: Humidity > 87%, CAPE > 750 $J\ kg^{-1}$, T < 27 °C	< 0.075 $mg\ m^{-3}$ > 0.075 $mg\ m^{-3}$	−0.123 0.198
Y11: Humidity < 87%, CAPE < 750 $J\ kg^{-1}$, T > 27 °C	< 0.085 $mg\ m^{-3}$ > 0.085 $mg\ m^{-3}$	0.045 −0.033
Y12: Humidity < 87%, CAPE > 750 $J\ kg^{-1}$, T > 27 °C	< 0.110 $mg\ m^{-3}$ > 0.110 $mg\ m^{-3}$	0.101 −0.049
Y13: Humidity > 87%, CAPE < 750 $J\ kg^{-1}$, T > 27 °C	< 0.070 $mg\ m^{-3}$ > 0.070 $mg\ m^{-3}$	−0.176 0.346
Y14: Humidity > 87%, CAPE > 750 $J\ kg^{-1}$, T > 27 °C	< 0.085 $mg\ m^{-3}$ > 0.085 $mg\ m^{-3}$	−0.026 0.010

(e.g., Lal et al., 2017; Penki and Kamra, 2013; Ramesh Kumar and Kamra, 2012) and are once again proof of the CAPE effect on lightning. Besides, it was found from Figs. 5 and 6 that CAPE plays an important role on the relation between aerosol and lightning.

It has long since been known that aerosol plays a role in convective activity. Aerosol particles influence the cloud microphysics by reducing the size of cloud droplet and increasing its number concentration, and then affect the vertical extent, lifetime, phase and optical properties of convective clouds, as well as lightning and precipitation of convective activity (Joos et al., 2017). However, the role of aerosol on convective activity is regulated by atmospheric humidity and other parameters.

Lal et al. (2013) have studied the relationship between cloud cover and aerosol over Indo-Gangetic Plain (IGP), India. They have found that in the low to moderate humid areas of IGP region, an increase in aerosol loading leads to an increase in High Cloud Cover (HCC) and a decrease in Low Cloud Cover (LCC); on the other hand, in high humid, an increase in aerosol loading does not show any significant effect on HCC, but LCC shows a positive trend. Lal et al. (2017) have studied the relationship between lightning and aerosol over IGP. They have found that lightning activity decreases (increases) with increasing aerosols during deficient (normal) monsoon rainfall years; besides, lightning increases with increasing aerosol during deficient rainfall years when the average value of AOD is < 0.88, thereafter lightning decreases. Then they have concluded that in low humidity area lightning increases with increasing aerosol for the low value of AOD, and decreases for the high value of AOD.

It was found in this study that lightning flash density has a significant positive correlation with PM_{10} when under dry condition ($R = 0.028$, $P = .048$, see in Table 4) rather than under wet condition. Besides, by comparing the peaks between lines $\times 3$ and $\times 4$ in Fig. 5b, between lines $\times 5$ and $\times 6$ in Fig. 5c, and between lines $\times 7$ and $\times 8$ in Fig. 5d, it was found that lightning flash density is higher for polluted cases than for clean cases when surface humidity is low. On the contrary, when surface humidity is high, lightning flash density is lower for polluted cases than for clean cases (line $\times 11$ versus line $\times 12$ in Fig. 5f, line $\times 13$ versus line $\times 14$ in Fig. 5g). Besides, for dry cases, lightning increases with increasing PM_{10} for the low value of PM_{10} , and decreases with increasing PM_{10} for the high value of PM_{10} (see lines Y1, Y3, Y4, Y7, Y8, Y11, Y12 in Fig. 6). While for wet cases, lightning decreases with increasing PM_{10} for the low value of PM_{10} , and increases with increasing PM_{10} for the high value of PM_{10} (see lines Y2, Y6, Y9, Y10, Y13, Y14 in Fig. 6). The results of this paper partly agree with the findings of Lal et al. (2013, 2017).

IPCC has reported (IPCC, 2007; Price, 2009) results from > 20 different models. These models have suggested that the greatest warming due to increasing greenhouse gases would occur not at the surface, but in the equatorial upper troposphere. If the upper atmosphere is warmed relative to the surface, the atmospheric lapse rate becomes more stable. So it is expect that the number of thunderstorms and the lightning activity would reduce due to stabilization of the atmosphere as the climate warms. However, all of these modeling indicate an approximate

10% increase in lightning activity globally for every 1 K global warming. Besides, studies on observation data also provided an apparent paradox. An opposite relationship between lightning and precipitation is found on the global scale for tropical continental centers of rain and lightning range in the opposite order, while Southeast Asia has less lightning during the wet La Nina years than during the drought EL Nino years.

With the data grouped by surface temperature, we have tried to gain some insights into these paradoxes. It was shown in Table 4 that surface temperature significantly correlates to lightning when CAPE is high; and CAPE has a more significant correlation with lightning when under cold condition than under warm condition; surface humidity has a significant correlation with lightning under warm conditions rather than under cold condition. For cold cases, the highest peak is found in the cluster high CAPE, high PM₁₀, low surface humidity (line × 4 in Fig. 5), and the strongest fluctuation of lightning following PM₁₀ is found in the cluster high CAPE, low surface humidity (line Y8 in Fig. 6). While for warm cases, the highest peak is found in the cluster low CAPE, low PM₁₀, high surface humidity (line × 13 in Fig. 5), and the strongest fluctuation of lightning following PM₁₀ is found in the cluster low CAPE, high surface humidity (line Y13 in Fig. 6). The physical reasons for this result are unknown. We presented the observations in the belief that it might stimulate further study and investigation.

5. Conclusions

Lightning and precipitation and their dependence on various parameters were studied in Nanning, China. It was found that not only the correlation between lightning and precipitation varies in different seasons, but also the dependence of lightning and precipitation on various parameters varies in different seasons. Then, quantitative analysis showed the joint effects of the surface temperature, surface humidity, CAPE and PM₁₀ on lightning. It was found that aerosol in the study area can change lightning activity. The aerosol effect is mainly regulated by humidity. In general, the correlation between aerosol and lightning for the dry cases is the opposite to that of the wet cases. CAPE and surface temperature could also affect the aerosol effect over lightning. However, in the present stage of knowledge, interactions of these effects, as well as their physical reasons, are too complex to draw a final conclusion, further precise observational and numerical studies are required.

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